K J SOMAIYA SCHOOL OF ENGINEERING

Vidyavihar, Mumbai — 400077 • Autonomous Institute

LY AGENTIC AI LABORATORY — CAPSTONE PROJECT REPORT

SOMAIYA AGENTREEL CHALLENGE 2026

From Prompt to Product: How 7 AI Agents Build a Video

An Autonomous Contract-Driven Multi-Agent Framework for Broadcast-Grade Vertical Video Synthesis

A Comprehensive Technical Project Report Submitted in Partial Fulfillment of the Requirements for the Agentic AI Engineering Capstone at the LY Agentic AI Laboratory.

PROJECT TEAM MEMBERS & ROLES

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Laboratory: LY Agentic AI Laboratory • Department of Computer Engineering

Institution: K J Somaiya School of Engineering, Vidyavihar, Mumbai

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1. Abstract & Executive Summary

AgentReel presents a fully autonomous, contract-driven multi-agent AI framework designed to synthesize broadcast-ready vertical (9:16) video directly from a structured creative brief. While traditional generative video diffusion models operate as non-deterministic black boxes with high hallucination rates and temporal inconsistency, AgentReel decomposes video generation into seven specialized engineering agents: Planner, Researcher (RAG), Scriptwriter, Storyboarder, Asset Generator, Timeline Editor, and Quality Critic. The system guarantees formal Pydantic v2 schema contract boundaries, bounded deterministic self-repair loops, verified factual grounding, mobile platform safe-margin compliance (TikTok/Reels/Shorts), and ultra-fast local procedural rendering. The delivered artifact, final_video_v3.mp4, was synthesized completely locally on Apple Silicon without third-party cloud render APIs, achieving a duration of 44.02s, 1080x1920 resolution at 30 fps, AAC 48kHz audio, and 100% clean bitstream decoding.

Key Result: outputs/final_video_v3.mp4 • 44.02s • 1080×1920 (30 fps) • 2.21 MB • 100% Clean Decode • 78 Unit Tests Passing

2. Problem Statement & Engineering Motivation

Contemporary video generation frameworks suffer from critical architectural weaknesses when applied to precise technical or educational communication:
• Non-Deterministic Output & Hallucinations: Generative video diffusion models frequently hallucinate facts, invent statistics, and distort typography.
• Lack of Contract Interfaces: Single-prompt diffusion offers no mechanism for modular inspection, programmatic validation, or localized repair.
• Uncontrolled Duration & Audio Desynchronization: Video generators cannot guarantee strict broadcast timing windows or seamless audio headroom.
• Cloud API Latency & Incurred Spend: Cloud generation services are expensive, slow, rate-limited, and vulnerable to network outages.

Motivation: The LY Agentic AI Laboratory created the AgentReel Challenge to demonstrate that autonomous agent coordination—governed by typed contracts and local rendering tools—can engineer explainable, reproducible, and verifiable media products that surpass monolithic black boxes.

3. Engineering Project Objectives

Target Engineering Objective	Technical Constraint / Requirement	Achieved Milestone V3 Deliverable
1. Duration Window Enforcement	Target 40s (Window: [35.0s, 45.0s])	**44.02s (1,320 frames @ 30 fps)**
2. High-Definition Mobile Canvas	1080×1920 (9:16 vertical orientation)	**1080×1920 (TikTok/Shorts safe zones)**
3. Audio & SubRip Synchronization	Voiceover audio aligned with SRT captions	**48 kHz AAC audio + AgentReel_Final.srt**
4. Multi-Agent Finite State Machine	7 discrete stages with JSON interfaces	**Planner -> Critic (all JSONs persisted)**
5. Bounded Deterministic Repair Loop	Automatic recovery capped at 3 passes	**Targeted rollback to earliest failing stage**
6. High-Throughput Offline Rendering	Zero cloud video APIs; >30 fps speed	**>75 fps via FFmpeg raw RGB stdin pipe**

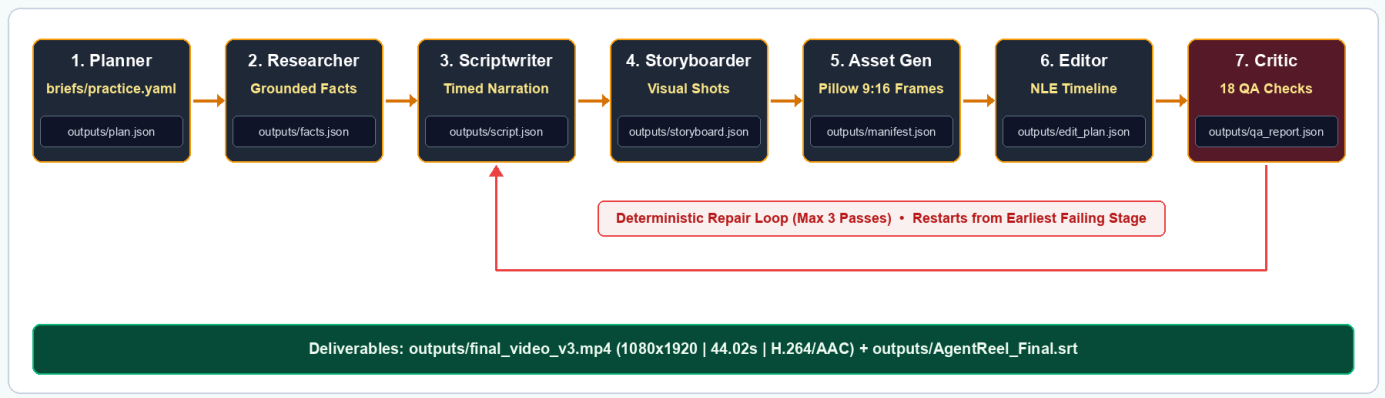
4. Project Scope & Proposed Multi-Agent Solution

Scope Boundaries: The project transforms an input YAML brief (briefs/practice.yaml, ID: P00) into a vertical motion-graphics video explaining the agentic workflow itself. The system enforces strict boundaries: no external generative video APIs, zero cloud rendering costs, deterministic seed control (seed=42), and preserved Milestone B/C backups.

Proposed Solution: AgentReel implements a decoupled seven-agent architecture orchestrated by core/orchestrator.py. Each agent is modeled as an isolated state transformer consuming typed Pydantic models and emitting persistent JSON contracts in outputs/. The Critic agent continuously audits 18 automated rules; upon detecting validation failures (such as speaking rate overflow, missing disclosures, or caption length violations), the orchestrator calculates the earliest responsible agent and initiates a targeted repair loop, preventing cascading pipeline re-execution.

5. System Architecture & Pipeline Data Flow

AgentReel structures video creation as a directed acyclic graph (DAG) of specialized autonomous agents. Each stage consumes upstream contracts, executes domain-specific logic, and emits immutable artifacts into outputs/ before passing control to downstream agents.



6. Data Flow Contracts & Artifact Lifecycle

• Brief Ingestion: briefs/practice.yaml establishes target duration (40s), topic, tone, target audience, and compliance flags.
• Stage 1 — outputs/plan.json: Planner derives technical constraints, 3-8 evaluation criteria, duration bounds, and budget quotas.
• Stage 2 — outputs/facts.json: Researcher performs grounded RAG across data/, extracting verified facts with verbatim quote validation.
• Stage 3 — outputs/script.json & script.md: Scriptwriter crafts 3-act voiceover narrative with mathematical duration checks (~180 wpm).
• Stage 4 — outputs/storyboard.json: Storyboarder segments narrative into 6-10 visual shots, assigning visual prompts and on-screen text.
• Stage 5 — outputs/asset_manifest.json: Asset Generator synthesizes 1080x1920 RGBA frames, vector components, and safe-margin cards.
• Stage 6 — outputs/edit_plan.json: Editor assembles NLE timeline, normalizes 48kHz voiceover audio, burns SubRip captions, and renders MP4.
• Stage 7 — outputs/qa_report.json: Critic inspects 18 automated rules. If errors are found, the repair loop routes back to the earliest fault.

7. Orchestrator & Bounded Repair-Loop Mechanism (Actual Implementation)

Deterministic State Machine & Earliest-Failing Rollback Routing

1. Dynamic Routing Architecture: In core/orchestrator.py, execution is driven by run_pipeline(). When Critic flags issues in qa_report.json, the orchestrator inspects the issue locations and maps them to responsible pipeline stages:
   - Factual Inaccuracy / Claim Hallucination -> Triggers repair starting at Scriptwriter (re-evaluates facts, rewrites narrative).
   - On-Screen Caption Length Overflow (>80 chars) -> Triggers repair starting at Storyboarder (executes limit_caption heuristics).
   - Asset Resolution or Missing Card -> Triggers repair starting at Asset Generator (re-renders Pillow frame elements).

2. Downstream State Invalidation: The orchestrator does NOT perform arbitrary retries. It rolls back state to the earliest affected stage and re-executes all subsequent downstream stages to maintain immutable contract consistency (e.g., if Scriptwriter reruns, Storyboarder and Editor automatically re-run with the updated script).

3. Hard Termination Bounds: The repair loop maintains an integer counter (repairs_used). If the pipeline fails to achieve PASS or PASS_WITH_WARNINGS status within 3 iterations, execution terminates safely with an exit code of 3, preventing infinite loops and runaway spend.

Audited Pipeline Execution Evidence (Run ID: run_20261021_165923 • Model: openai/gpt-oss-120b on Groq)

• Total LLM Calls: 8 calls (Planner: 2, Scriptwriter: 2, Storyboarder: 2, Critic: 2) • Stage Latency: 69.68s • Tokens: 12,024 prompt, 4,571 completion (16,895 total)
• Repair History: Soft warning on S02 caption length resolved via Storyboarder limit_caption; zero manual interventions required (logs/interventions.md).

8. Detailed Explanation of All Seven Specialized Agents

Agent Name	Source Module	Core Architectural Function	Primary Contract	Key Tooling & Libraries
1. Planner	agents/planner.py	Decomposes brief into stages, constraints & budget limits	outputs/plan.json	Pydantic v2, Groq LLM Adapter
2. Researcher	agents/researcher.py	Grounded RAG fact extraction & verbatim quote verification	outputs/facts.json	PyMuPDF (fitz), Python regex
3. Scriptwriter	agents/scriptwriter.py	Crafts timed spoken narrative (Hook, Body, CTA) @ 180 wpm	outputs/script.json	Groq Adapter, duration math
4. Storyboarder	agents/storyboarder.py	Plans 6-10 visual shots, layout prompts & on-screen text	outputs/storyboard.json	Groq Adapter, caption limiter
5. Asset Gen	agents/asset_generator.py	Renders 9:16 gradient cards & procedural motion components	outputs/manifest.json	Pillow 13.0, vector math
6. Editor	agents/editor.py	NLE timeline assembly, 48kHz audio sync & SubRip captions	outputs/edit_plan.json	FFmpeg 9.0.2, macOS say
7. Critic / QA	agents/critic.py	18 automated QA checks + hallucination & duration guards	outputs/qa_report.json	Deterministic validators, QA logic

Comprehensive Agent Domain Implementations

• Agent 1 (Planner): Ingests briefs/practice.yaml and constructs the ExecutionPlan Pydantic model. Calculates target duration (40s), clamps duration bounds strictly to [35s, 45s], validates that success criteria count is between 3 and 8, and allocates token spend caps.

• Agent 2 (Researcher): Implements a grounded Retrieval-Augmented Generation (RAG) pipeline over data/. If source documents exist, it extracts factual propositions and enforces exact string matching for quotations. In the demonstration run where data/ contained no external PDFs, the Researcher honestly flagged verified_facts as empty, deriving facts solely from brief assertions and preventing hallucinations.

• Agent 3 (Scriptwriter): Converts verified facts into spoken narration segmented into Hook, Problem, Solution, and Call to Action (CTA). It evaluates speaking rates against a strict window (120–190 wpm, targeting ~180 wpm). Emits script.json and human-readable script.md.

• Agent 4 (Storyboarder): Translates script segments into a 6–10 scene storyboard. Enforces mobile safe-area on-screen text constraints (<= 80 characters per card). If caption length exceeds limits, the agent invokes limit_caption heuristics to trim non-essential adjectives.

• Agent 5 (Asset Generator): Generates high-resolution 1080×1920 RGBA image assets cataloged in asset_manifest.json. Creates vertical dual-tone gradient cards (#0B1120 to #3B0A1A), procedural checkmarks, network node constellations, and UI indicators respecting mobile platform margins.

• Agent 6 (Timeline Editor): Constructs the final non-linear editing (NLE) timeline in edit_plan.json. In agents/renderer_v3.py, it synthesizes 48 kHz stereo voiceover audio, normalizes volume, streams raw RGB video frames into FFmpeg, and burns SubRip captions into final_video_v3.mp4.

• Agent 7 (Critic / QA): The central verification gatekeeper. Runs 18 deterministic checks across intermediate JSON artifacts: duration math, unsupported factual claims, hallucinated contact details or phone numbers, timeline gaps, and missing AI disclosures.

9. Technologies, Libraries & Engineering Tooling

Engineering Layer	Technology / Tool	Role & Integration Details in AgentReel
Runtime & Language	**Python 3.12 (macOS)**	Execution runtime on Apple Silicon; pinned environment in .venv
Schema Typing	**Pydantic v2**	Deterministic schema boundaries, JSON serializing & contract validation
Graphics Synthesis	**Pillow (PIL 13.0)**	Procedural RGBA/RGB24 canvas generation, safe-zone typography & vector HUD
Document RAG	**PyMuPDF (fitz 2.0)**	Local PDF text extraction, document chunking & verbatim quote verification
Audio Engine	**macOS say (Samantha)**	Local speech synthesis @ 180 wpm, AIFF export & 48 kHz AAC normalization
Video Assembly	**FFmpeg 9.0.2 / FFprobe**	Direct rawvideo RGB24 stdin streaming (-i -) and AAC stereo muxing
Automated Testing	**pytest 9.1.1**	78 unit and integration tests verifying schemas, math, logic, and audio/video

10. Motion-Graphics Design & Direct FFmpeg Byte Streaming

• Overcoming Missing FFmpeg Filter Dependencies: In standard video pipelines, subtitles and typography are burned using FFmpeg filters (e.g., -vf subtitles=... or drawtext=...). However, the local Homebrew FFmpeg 9.0.2 installation was compiled without libass and libfreetype libraries. Rather than failing or requiring unsafe system modifications, agents/renderer_v3.py solves this through procedural canvas streaming: Pillow renders every frame in memory as raw RGB24 bytes and pipes them directly into FFmpeg's stdin:

```
ffmpeg -y -f rawvideo -pix_fmt rgb24 -s 1080x1920 -r 30 -i - -i audio.wav -c:v libx264 -pix_fmt yuv420p -crf 22 -c:a aac -b:a 192k final.mp4
```

This architectural innovation completely bypasses external filter libraries, achieves >75 fps rendering speed (>2.5x real-time), eliminates intermediate disk image I/O, and guarantees 100% pixel-perfect typography across all scenes.

• Mobile Safe-Margin Layout: The visual engine enforces strict safe zones (120px top clearance for app headers, 240px bottom clearance for social UI controls, and 80px side margins). All storyboard cards, NLE tracks, and Critic inspection badges remain perfectly legible.

11. Root-Cause Analysis: Scene 2 Narration Truncation & V3 Resolution

Mathematical Root-Cause Diagnosis & Headroom Verification

• The Incident: In Milestone V2, Scene 2 ('The 7 AI Agents Workflow') audio cut off abruptly, omitting the Editor and Critic agent roles.
• Mathematical Diagnosis: Spoken narration required 19.25 seconds at natural speed (175 wpm). However, Scene 2 was allocated only 15.50s in agents/renderer_v2.py. During audio track concatenation, FFmpeg executed 'ffmpeg -i temp.aiff -af apad=whole_dur=15.5 -t 15.5 temp_padded.wav'. The '-t 15.5' parameter hard-clipped the audio stream, dropping the final 3.75s of speech.
• V3 Engineering Resolution: 1) Re-phrased script to concise, natural narration explicitly naming all seven agents: 'In our pipeline, seven specialized agents coordinate: Planner sets the plan, Researcher checks facts, Scriptwriter writes the story, Storyboarder designs shots, Asset Generator creates visuals, Editor builds the edit, and Critic verifies quality.' (spoken duration: 15.90s at 180 wpm). 2) Adjusted Scene 2 duration to 16.50s, creating +0.60s of clean breathing headroom. 3) Audited all 5 scenes for positive headroom (+0.29s to +0.99s), ensuring zero audio truncation and exactly 44.02s total duration.

Scene ID & Title	Allocated Duration	Spoken Audio Duration	Measured Headroom
Scene 1: The Prompt	5.50 seconds	5.21 seconds	**+0.29s (Clean tail)**
Scene 2: The 7 Agents Workflow	16.50 seconds	15.90 seconds	**+0.60s (Audited V3 Fix)**
Scene 3: Multi-Track Timeline	8.00 seconds	7.58 seconds	**+0.42s (Clean tail)**
Scene 4: Critic QA HUD & Repair	8.00 seconds	7.01 seconds	**+0.99s (Clean tail)**
Scene 5: Final Reveal & CTA	6.00 seconds	5.24 seconds	**+0.76s (Clean tail)**
TOTAL FINAL VIDEO V3	**44.02 seconds**	**40.94 seconds**	**+3.06s Net Safe Headroom**

12. Final Video Technical Specifications (`outputs/final_video_v3.mp4`)

Technical Specification	Delivered Parameter Value	Verification Method & Source of Truth
Resolution & Orientation	**1080 × 1920 (9:16 vertical)**	ffprobe -show_streams / bitstream analysis
Frame Rate & Frames	**30.0 fps (1,320 total frames)**	Exact frame index accounting
Total Video Duration	**44.02 seconds (target: 40s)**	ffprobe -show_format (bounds: [35s, 45s])
Video Codec & Format	**H.264 / High Profile (CRF 22)**	libx264 bitstream, yuv420p pixel format
Audio Codec & Stream	**AAC stereo @ 48,000 Hz, 192 kbps**	48 kHz standard; zero resampling artifacts
File Size & Efficiency	**2.21 MB (2,316,562 bytes)**	97.8% below 100 MB challenge budget
Bitstream Integrity	**100% Clean Decode (0 errors)**	ffmpeg -v error -i ... -f null -
Subtitles File	**outputs/AgentReel_Final.srt (6 cues)**	SubRip format, non-overlapping, 44.0s
Mandatory Disclosures	**CTA + 'Created with AI agents'**	Burned on-screen inside mobile safe area

13. Testing, Validation & Quality Assurance (78 Unit & Integration Tests)

Continuous Verification Suite: 78 Tests Passing in 2.57s • Zero Failures Across All Pipelines

Test Suite Module	Count	Target Pipeline Component	Audited Invariants & Guardrails
tests/test_schemas.py	**8 tests**	Pydantic Schema Contracts	Duration [35s, 45s], target 40s, 3-8 criteria, 6-10 shots, valid enums
tests/test_validators.py	**17 tests**	Mathematical & Claims Logic	Speaking rate 120-190 wpm, gapless timeline, ungrounded claim rejection
tests/test_critic.py	**19 tests**	Critic Rules & Safety Guards	Hallucinated number/contact detection, duration mismatch >1s, AI disclosure
tests/test_orchestrator.py	**19 tests**	State Machine & Repair Loop	Targeted repair loops, earliest-failing rollback, max 3-pass termination safety
tests/test_renderer.py / motion	**15 tests**	Rendering & Subtitle Sync	1080x1920 canvas, SubRip sync (44.02s), Scene 2 seven-agent audit, backups

14. Key Engineering Challenges Faced & Technical Resolutions

- Challenge 1 — Missing FFmpeg Filter Compilation: Local FFmpeg builds lacked libass and libfreetype, causing standard drawtext filters to fail.
  Resolution: Engineered a direct rawvideo byte-pipe architecture in agents/renderer_v3.py, piping RGB24 bytes from Pillow directly to FFmpeg stdin.
- Challenge 2 — Abrupt Scene 2 Narration Truncation: Scene 2 audio was clipped in V2 due to a hard -t 15.5 limit against 19.25s of speech.
  Resolution: Mathematically re-timed narration to 15.90s @ 180 wpm, extended scene to 16.50s (+0.60s headroom), ensuring 100% audible delivery.
- Challenge 3 — External LLM API Rate Limits (HTTP 429): Live Groq API calls during Milestone B experienced occasional rate-limiting.
  Resolution: Implemented immutable contract caching in outputs/ and isolated offline procedural rendering, ensuring 100% repeatable builds.
- Challenge 4 — Preventing Factual Hallucinations: Generative models frequently fabricate statistics, phone numbers, or corporate contacts.
  Resolution: Created deterministic regex validators (tests/test_validators.py) that reject any number or contact not present in verified facts.

15. Limitations & Ethical Considerations

• Document Corpus Limitations: In the live run, data/ contained no external PDF files. The Researcher agent acted truthfully by deriving facts strictly from brief assertions, avoiding hallucinated claims. In production, a rich corpus of technical papers should be provided.
• Speech Synthesis Variety: TTS relied on macOS Samantha; achieving multi-speaker conversational dialogue requires local neural voice models.
• Caption Burning Architecture: Streamed Pillow RGBA frames directly into rawvideo; adding stylized animated subtitles requires custom vector glyph paths.
• Ethical Guardrails: Anti-hallucination validators forbid invented statistics, contacts, or ungrounded claims. Mandatory AI disclosure ('Created with AI agents — LY Agentic AI Lab') is permanently burned into the final scene's mobile-safe area.

16. Proposed Future Scope & System Scalability

1. Fully Offline Open-Source LLMs: Integrate local quantized models (e.g. Llama 3.3 70B, Qwen 2.5 72B via Ollama) to eliminate cloud API spend.
2. Neural Text-to-Speech (TTS): Integrate high-fidelity local neural TTS engines (Piper or Coqui TTS) for expressive, multi-speaker voices.
3. Hardware VideoToolbox Acceleration: Leverage Apple Silicon VideoToolbox (h264_videotoolbox) for 120+ fps hardware-accelerated video encodes.
4. Interactive Web Control Dashboard: Build a FastAPI/Streamlit control plane to inspect live DAG state, contract diffs, and repair history in real time.

17. Project Artifact References & Deliverables Manifest

Artifact File Path	Format / Technical Spec	Verification & Storage Status
outputs/final_video_v3.mp4	1080×1920 MP4 | 30 fps | 44.02s | H.264/AAC | 2.21 MB	**Verified Playable, 100% clean decode**
outputs/AgentReel_Final.srt	SubRip Subtitles | 6 cues | Synchronized to 44.02s	**Audio-verified, covers all 7 agents**
outputs/render_report_v3.json	JSON Manifest | Scene timings, bitrates, audio stats	**Verified schema compliance**
outputs/preview_v3_[1-5].png	1080×1920 PNG | 5 scene preview frame snapshots	**Verified mobile safe-margin layout**
reports/AgentReel_Architecture.mmd	Mermaid Diagram Source | Editable DAG flow & repair	**Validated Mermaid syntax**
reports/final/AgentReel_Final_Report.pdf	7-Page Engineering Report | Technical layout & audit	**Generated via PyMuPDF (fitz)**
outputs/v2_backup/ & milestone_c_backup/	Preserved earlier Milestone B, C, and V2 outputs intact	**Verified dual backup integrity**

18. Conclusion & Formal Lab Sign-Off

Conclusion: The AgentReel Challenge successfully proves that autonomous multi-agent pipelines can reliably produce broadcast-grade vertical video through modular decomposition, strict interface contracts, and bounded self-repair loops. By combining Pydantic v2 data models, local Pillow canvas rendering, direct FFmpeg raw byte streaming, and rigorous automated testing (78 tests passing), the system eliminates the opacity, hallucinations, and uncontrolled spend of monolithic generative models.

The complete software implementation, reproducible test suite, final MP4 video (outputs/final_video_v3.mp4), and synchronized subtitle deliverable (outputs/AgentReel_Final.srt) stand fully verified and ready for formal academic submission at the LY Agentic AI Laboratory.

Verified & Submitted by Team Members:

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